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What Are the Requirements for Design Thinking Articles?

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Abstract: This paper investigated how design thinking is researched in management studies. A review of articles published in major business management journals found commonalities in that (a) while design thinking consists of several studies and includes various elements, (b) the most recent empirical research has in common that based on Brown (2009) and Martin (2009), along with practices and tools, the themes of user centeredness and experimentation are discussed.

Keywords: design thinking, product development, design management, review

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Introduction

Design thinking is a well-known methodology adopted by the design consulting firm IDEO (Brown, 2008). According to Brown (2008), which is a representative paper on this topic, design thinking is "a methodology that imbues the full spectrum of innovation activities with a human-centered design ethos" (Brown, 2008, p. 86), and it is a way of thinking that provides a framework for the process of product development and the activities within it.[1]

Brown (2008) asserts that it is necessary to repeat the process of (i) inspiration, (ii) ideation, and (iii) implementation. He also proposes taking a human-centered approach, repeating challenging activities, and utilizing human resources from different fields. Brown (2008) and Brown (2009) which is based on Brown (2008) describe an example of applying design thinking by using the case of joint bicycle development by the Japanese bicycle maker Shimano, Inc., and IDEO. IDEO's interdisciplinary team discussed project challenges with Shimano and then (i) they interviewed many types of bicycle users about why people in the US did not ride bicycles and derived problems such as maintenance problems (inspiration), (ii) the designers suggest a concept called coasting, in which bicycles do not have brakes or gears (ideation), and (iii) they came up with sales strategies (implementation).[2]

Although design thinking is linked to management, it has by no means been adequately discussed in the field of academic management studies. Therefore, this paper reviewed articles on design thinking. The results showed that (a) while existing review

[1] Design management has become popular also in Japan in recent years (e.g., Akiike, 2017; Akiike & Yoshioka-Kobayashi, 2017; Akiike, Yoshioka-Kobayashi, & Katsumata, 2019; Hanahara, 2021).

[2] (i), (ii), and (iii) are the authors' classification. Brown (2008, 2009) himself does not explicitly classify these into (i), (ii), and (iii).

articles pointed out the diversity of the elements involved in design thinking, (b) empirical research about design thinking were based on Brown (2008, 2009) and Martin (2009), and addressed the common themes of user center.

Method

This paper uses the Web of Science, a database of academic papers, to search for articles on design thinking in management field

Table 1. List of design thinking articles in the management studies

	type	journal
Johansson-Sköldberg et al. (2013)	review	CIM
Seidel & Fixson (2013)	qualitative	JPIM
Liedtka (2015)	review	JPIM
Carlgren, Elmquist, & Rauth (2016)	qualitative	CIM
Carlgren, Rauth, & Elmquist (2016)	qualitative	CIM
Elsbach & Stigliani (2018)	review	Journal of Management
Kummitha (2019)	qualitative	CIM
Micheli et al (2019)	review	JPIM
Coco, Calcagno, & Lusiani (2020)	qualitative	CIM
Da Silva, Kaminski, & Armellini (2020)	qualitative	CIM
Dell'Era et al. (2020)	qualitative	CIM
Nagaraj et al. (2020)	quantitative	JPIM
Roth, Globocnik, Rau, & Neyer (2020)	quantitative	CIM
Magistretti, Dell'Era, Verganti, & Bianchi (2021)	qualitative	R&D Management
Starostka, Evald, Clarke, & Hansen (2021)	qualitative	CIM
Klenner et al (2021)	qualitative	JPIM
Magistretti, Ardito, & Messeni Petruzzelli (2021)	review	JPIM

Note: For the review paper, the authors checked the contents and made decisions.
Source: Web of Science

according to the following three criteria: (1) journals selected based on Bruton and Lau (2008) and Perks and Roberts (2013): (2) articles with "design thinking" contained in the topic and title; and (3) document type of either "article" or "review."[3]

The search found 17 relevant papers (Table 1). The article types consisted of five review articles, 10 qualitative articles, and two quantitative articles. Nine articles were from *Creativity and Innovation Management* (CIM); six were from the *Journal of Product Innovation Management* (JPIM); one was from the *Journal of Management*; and one was from *R&D Management*. Moreover, 11 of the articles were published in 2018 or later, which indicates that design thinking has become emerging theme in management studies. This paper examines the impact on subsequent empirical research based on the points raised in review articles on the theme of design thinking in the field of management studies.

Review of Design Thinking in Management Studies

First, this paper describes four representative review articles.[4] Johansson-Sköldberg, Woodilla, and Çetinkaya (2013) examine the genealogy of design thinking, so we will use it as the basis about design thinking. According to Johansson-Sköldberg et al. (2013), current research papers on design thinking are rooted in two approaches: designerly thinking, which examines the designer mentality cultivated in design schools, and design thinking, which was originated by IDEO. Johansson-Sköldberg et al. (2013) pointed out that the researchers discussed these independently.[5] First we

[3] As of August 9, 2021.

[4] The number of citations in the Web of Science was 257, 144, 65, and 70, respectively.

[5] It should be noted that the concept of designerly thinking does not appear in Simon (1969, as cited in Johansson-Sköldberg et al., 2013) although this was stated in Johansson-Sköldberg et al. (2013). Schön (2016),

introduce design thinking. It is simplified for non-designers, and there are three types of core research (Johansson-Sköldberg et al., 2013).[6]

(A) Brown (2008) discussed design thinking based on IDEO's mindset. (B) The core research of Martin (2009) and others advocated "a way to approach indeterminate organizational problems, and a necessary skill for practicing managers" (Johansson-Sköldberg et al., 2013, p. 128). (C) The core research by Boland and Collapy (2004) held that design thinking is part of management theory (Johansson-Sköldberg et al., 2013).

Johansson-Sköldberg et al. (2013) argue that the other type of designerly thinking originated with Simon (1969, as cited in Johansson-Sköldberg et al., 2013). There are five types of core research. (a) Simon views the process of considering artifact design as a problem-solving exercise and focuses on the exploratory activities involved (Simon, 1996).[7] (b) Other core research includes Schön (2016), which focuses on reflections on the professional activities of the designer,[8] (c) Buchanan (1992), who focused on problem solving, (d) core research such as Lawson (1990), which focused on the interpretation of such things as framing, reasoning, and imaging,[9] and (e) Krippendolf (2006), which focused on products' meanings (Johansson-Sköldberg et al., 2013). This situation shows that design thinking consists of not only Brown (2008, 2009) but also

Buchanan (1992), and Lawson (1990) call it "design thinking."

6 Johansson-Sköldberg et al. (2013) call this "discourse," but this paper calls it "core research."

7 Johansson-Sköldberg et al. (2013) cite both the first (1969) and third (1996) editions of Simon, but we were unable to verify the 1969 edition.

8 Although the authors succeeded in verifying Schön (2016), Johannsson-Sköldberg et al. (2013) cited the first edition, which was published in 1983 (Schön, 1983, as cited in Johansson-Sköldberg et. al., 2013).

9 Johansson-Sköldberg et al. (2013) list the first edition from 1980 in the bibliography but quote the 4th edition published in 2006 (Lawson, 1980, 2006, as cited in Johansson-Sköldberg et al., 2013). The authors were able to confirm Lawson (1990).

several studies.

Liedtka (2015) uses a review of prior research to investigate how design thinking has the effect of reducing psychological bias. The description in the previous research review section of Liedtka (2015) heavily focuses on practices and tools such as the process of design thinking and brainstorming so on.

Elsbach and Stigliani (2018) conduct review of articles and conceptualization of the relationship between design thinking and organizational culture. Elsbach and Stigliani (2018) describes how design thinking affects organizational culture during the four stages of (I) experiences and feedback, (II) reflection on experiences, (III) develop general theories explaining experiences, and (IV) test general theory. It is common to Liedtka (2015) in that Elsbach and Stigliani (2018) discuss the relationships with the tools used in design thinking at each stage.

Finally, Micheli, Wilner, Bhatti, Mura, and Beverland (2019) review the previous literatures to derive the theme which is mentioned in many studies with citing Johansson-Sköldberg et al. (2013), Liedtka (2015), Elsbach and Stigliani (2018). This resulted in clarification that the themes of discussion were creativity and innovation, user centeredness and involvement, problem solving, iteration and experimentation, interdisciplinary collaboration, ability to visualize, gestalt view, abductive reasoning, tolerance of ambiguity and failure, blending rationality and intuition, and design tools and methods (Micheli et al., 2019, Table 3, pp. 132–133).

From these review articles, it is clear that current research on design thinking contains many types of core research and consists of various elements. It is therefore necessary to consider which elements to take up when one writes an article on design thinking in management studies. Additionally, all of these articles focus on “tools.” It is one characteristic of research on design thinking in the field of management studies (Morinaga, 2020).

Design Thinking in Empirical Management Studies

These review articles do not cover only articles about design thinking in management studies when they review. We therefore surveyed empirical papers in management studies published after Micheli et al. (2019) for understanding how they treated the elements of design thinking.

From a variety of journals and analytical methods, we selected three articles: Dell'Era, Magistretti, Cautela, Verganti, and Zurlo (2020), Nagaraj, Berente, Lyytinen, and Gaskin (2020), and Magistretti, Dell'Era, Verganti, and Bianchi (2021).

Dell'Era et al. (2020) conduct qualitative study. It organizes the elements of design thinking into human-centered design, problem framing, diversity, and experimentation, based on Micheli et al. (2019) and other prior studies. Based on this classification, a case-study analysis of the understanding and introduction of design thinking in consulting organizations resulted in classifying design thinking into four types: creative problem solving, sprint execution, creative confidence, and innovation of meaning. In this discussion, they describe such practices and tools as brainstorming and the naïve mind.

Nagaraj et al. (2020) conduct qualitative study that builds up the concept of team design thinking while focusing on the use of design thinking by teams. Citing Micheli et al. (2019) and in reexamining previous studies on design thinking, Nagaraj et al. (2020) organize design thinking into the four elements of user empathy, collaborative abduction, iteration, and collaborative representation and created a questionnaire form using them.[10] In addition, some questionnaire

[10] Nagaraj et al. (2020) describe Micheli et al. (2019) as Micheli et al. (2018), but they are both the same. The element of experimentation also appears in the iterations classified by Nagaraj et al. (2020), so it is considered to be almost the same element as the experimentation derived by Dell'Era et

items include such tools as prototypes and brainstorming. Nagaraj et al. (2020) collected data and analyzed the impact of design thinking on product utility and novelty by using product unfamiliarity as a moderator. The analysis found that design thinking has a positive impact on product utility regardless of product unfamiliarity, but it has a positive impact on product novelty only when there is familiarity with the product (Nagaraj et al., 2020).

Magistretti, Dell'Era, Verganti, and Bianchi (2021) conduct qualitative study including content analysis which analyzes the introducing process of the way of thinking "Proxemics" in firms that practice it.

Although it focuses on the themes of user centeredness and experimentation, Magistretti, Dell'Era, Verganti, and Bianchi (2021) point out that the logic and methods of design thinking differ between development and research process.

For example, even for the same element of user centeredness, the development process requires exploration of needs from the standpoint of the user (empathy) using methods such as ethnography, while the research process requires abstract conceptualization of the user using tools such as interaction maps in Proxiemics (Magistretti, Dell'Era, Verganti, and Bianchi, 2021). The article also discusses tool such as prototype.

Requirements for Design Thinking Articles in Management Studies

In Dell'Era et al. (2020), Nagaraj et al. (2020), and Magistretti, Dell'Era, Verganti, and Bianchi (2021), there are slight variations in the elements of design thinking. Although all three papers cite Micheli et al. (2019), they reorganize the elements of design thinking. This is because that Micheli et al. (2019) only presented a

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al. (2020).

list of themes addressed in articles about design thinking.

Although design thinking has become jungle by absorbing various fields, the themes offered by Micheli et al. (2019) have a clear hierarchy of logical relationships. For example, when we compare the themes of creativity and innovation with the theme of abductive reasoning in Table 3 of Micheli et al. (2019, p. 132), we can infer the former might be higher-level theme as a logical relationship. Therefore, future research will need to elaborate on the logical relationship of these themes.

At the same time, these three papers have three common traits relating to the theme of design thinking: (1) they treated the recursive and repetitive processes such as iteration and experimentation; (2) they indicated the need to focus on the user; and (3) they examined both practices and tools that can contribute to actual design activities. Moreover, Brown (2009) and Martin (2009) have already identified these three characteristics. As a matter of fact, these two literatures are cited in all the above articles (four review articles and three empirical studies). Considering this, Brown (2008, 2009) and Martin (2009) can be regarded as indispensable literatures when addressing design thinking in management studies.

Conclusion

This paper reviewed management studies articles that discuss design thinking, which is the framework for product development. The conclusion is that: (a) while the review articles indicate that design thinking is a concept made up of various elements, and the elements of design thinking differ in empirical research, (b) it was common to discuss experimentation and user centered approach together with practice and tools by citing Brown (2009) and Martin (2009). If you want to write a design thinking paper as management studies, you need to follow the above points with citing Brown (2009) and Martin (2009).

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